

I. NUCLEOSYNTHESIS:

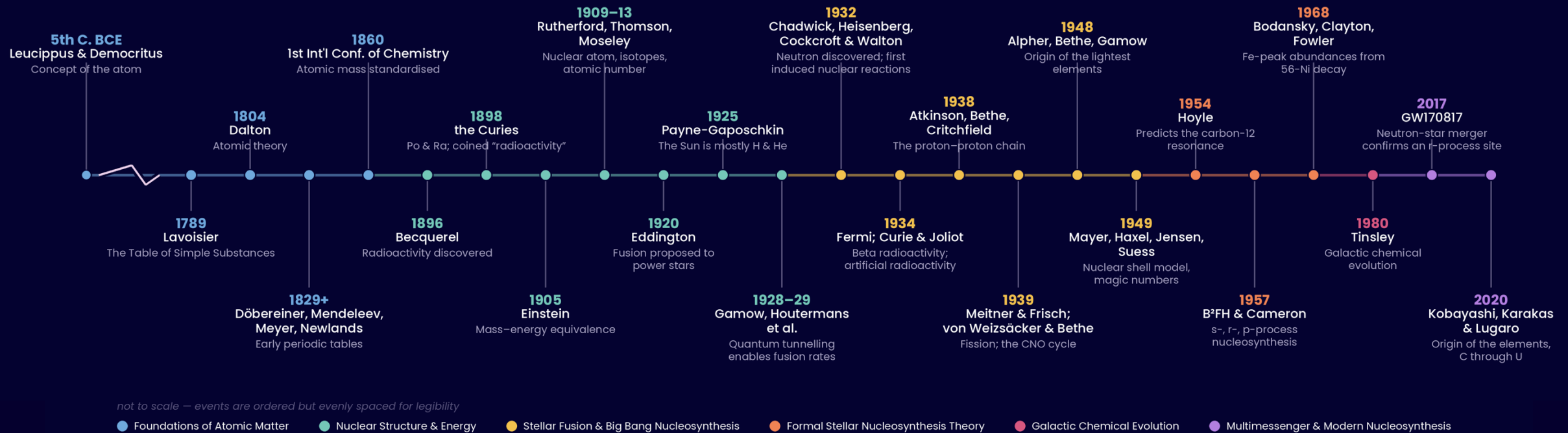
The stellar origins of the elements

Dr Giulia Cinquegrana

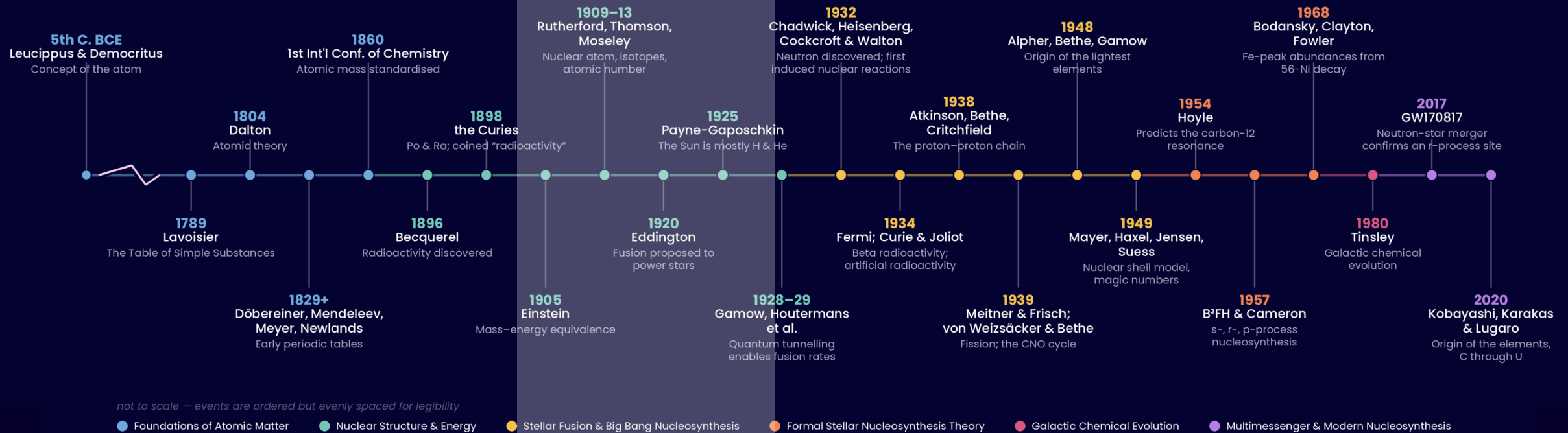
Swinburne University of Technology, Melbourne



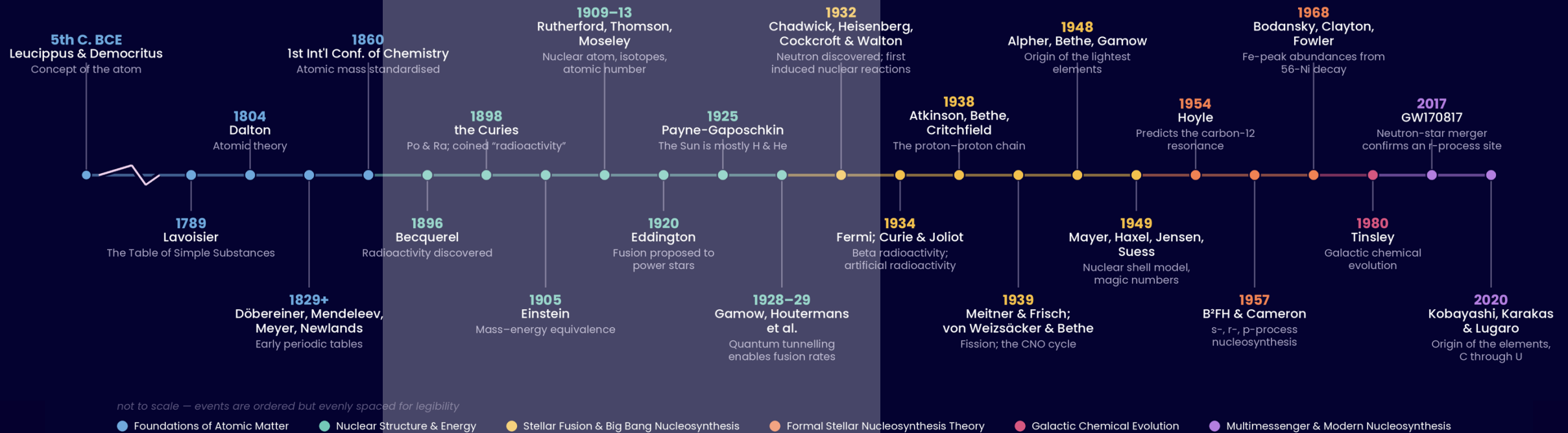
A Brief History of Nucleosynthesis



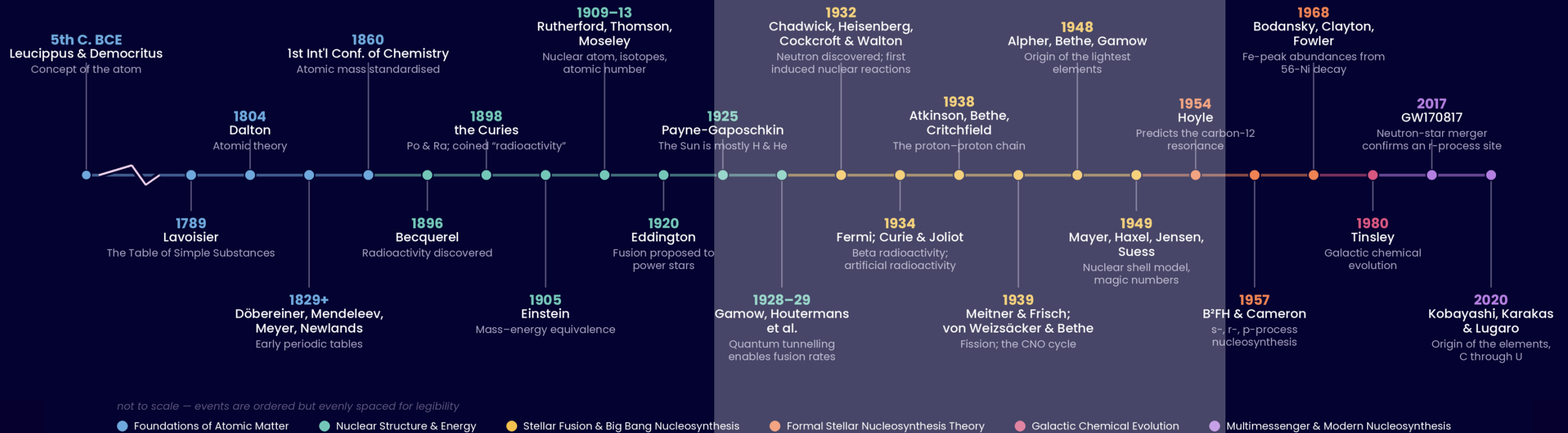
A Brief History of Nucleosynthesis



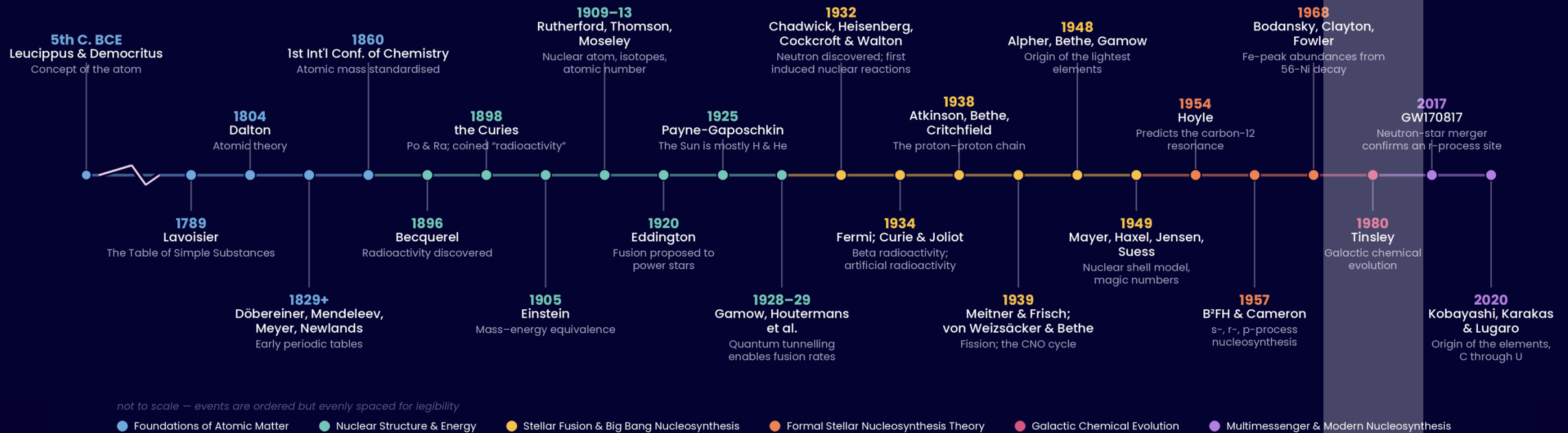
A Brief History of Nucleosynthesis



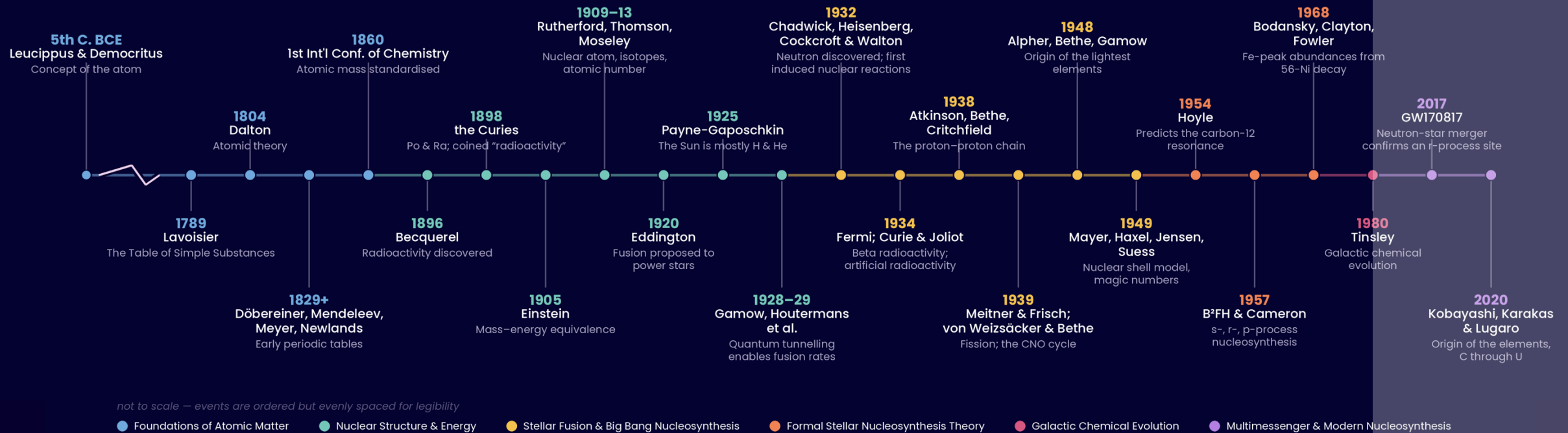
A Brief History of Nucleosynthesis



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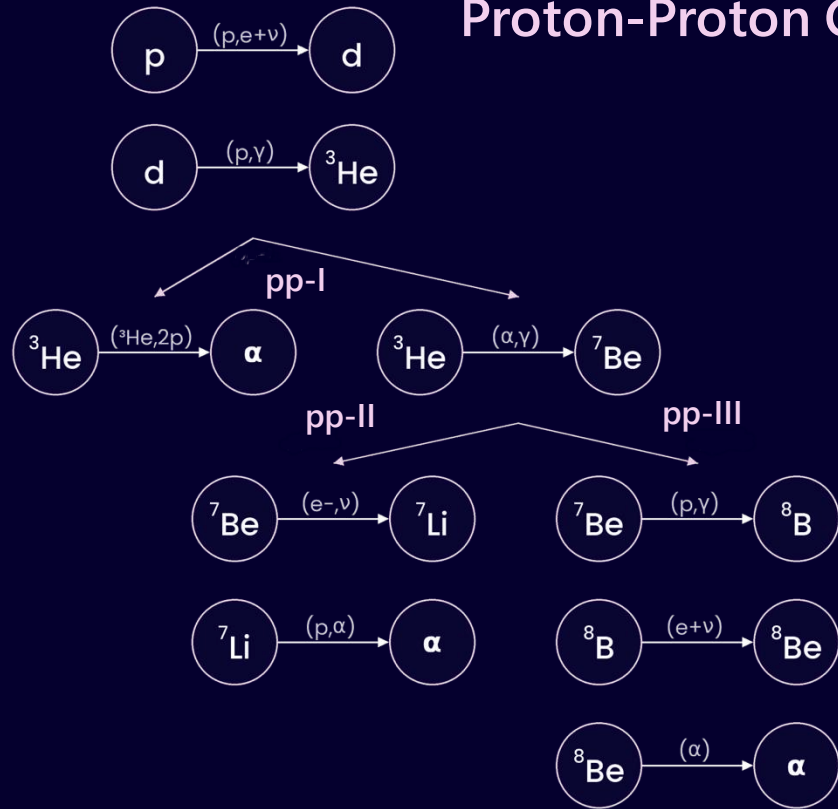


A Brief History of Nucleosynthesis



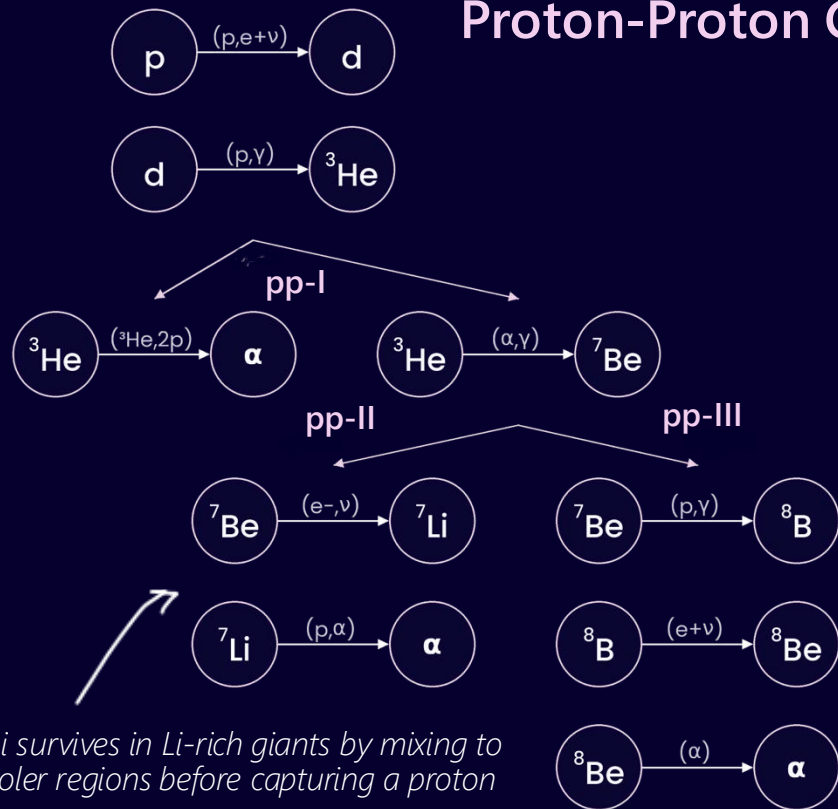
Hydrogen burning

Proton-Proton Chains



Hydrogen burning

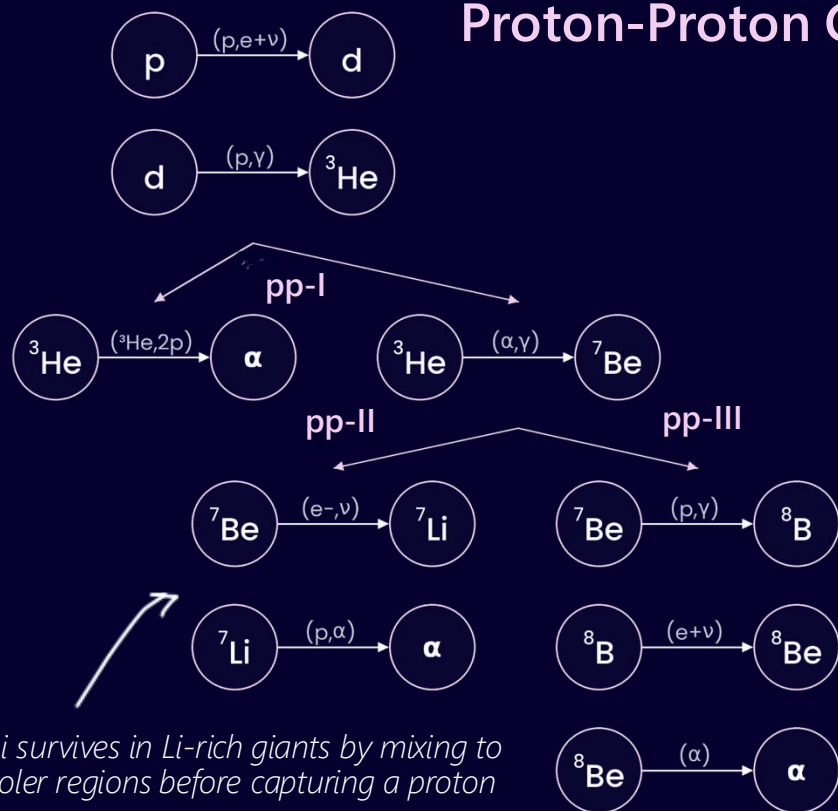
Proton-Proton Chains



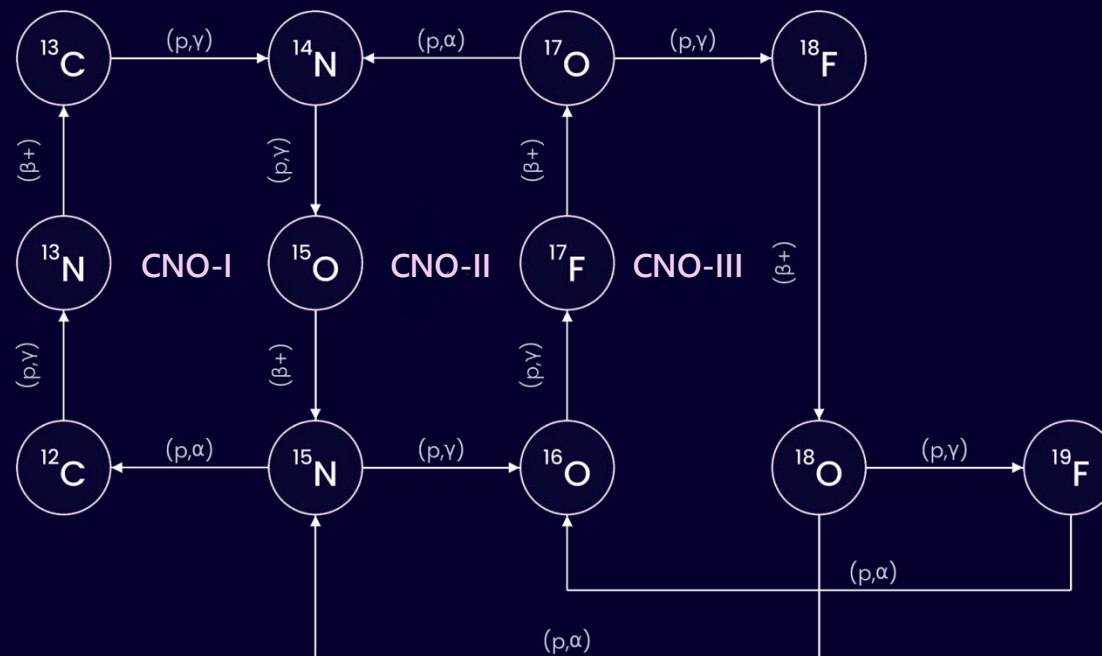
** ${}^7\text{Li}$ survives in Li-rich giants by mixing to cooler regions before capturing a proton*

Hydrogen burning

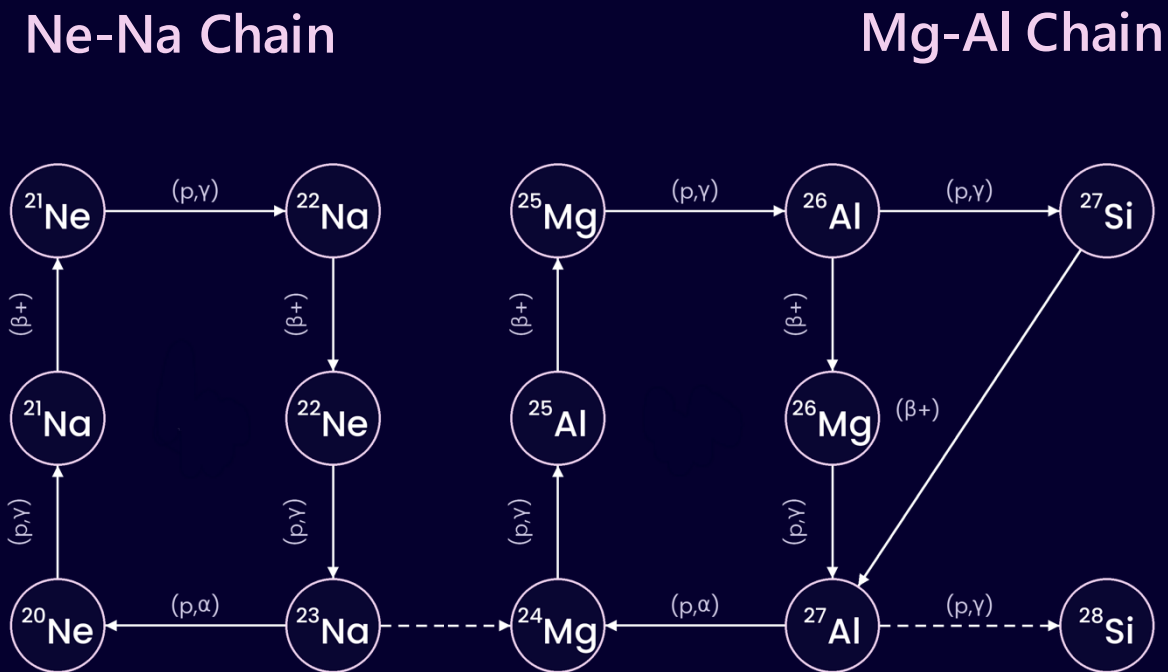
Proton-Proton Chains



CNO Cycles



Hydrogen burning



**Ne-Na and Mg-Al chains in AGBs are responsible for the light-element anticorrelations observed in globular clusters.*

He, C and O burning

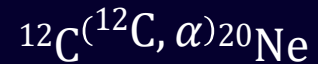
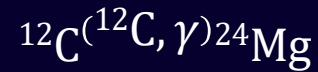
He burning, $T_c \gtrsim 10^8 \text{ K}$



Once there is sufficient reservoir of ^{12}C ,



C burning, $T_c \gtrsim 5 \times 10^8 \text{ K}$



} most probable

O burning, $T_c \gtrsim 1.5 \times 10^9 \text{ K}$

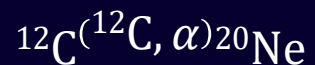
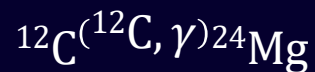


} most probable

He, C and O burning

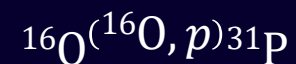
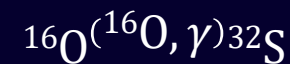
Chain	Q
$4p \rightarrow \alpha$	26.8
$3\alpha \rightarrow {}^{12}\text{C}$	7.3
${}^{12}\text{C}({}^{12}\text{C}, \alpha){}^{20}\text{Ne}$	4.6
${}^{12}\text{C}({}^{12}\text{C}, p){}^{23}\text{Na}$	2.2
${}^{12}\text{C}({}^{12}\text{C}, \gamma){}^{24}\text{Mg}$	13.9
${}^{12}\text{C}({}^{12}\text{C}, 2\alpha){}^{16}\text{O}$	-0.1
${}^{12}\text{C}({}^{12}\text{C}, n){}^{23}\text{Mg}$	-2.0
${}^{16}\text{O}({}^{16}\text{O}, \alpha){}^{28}\text{Si}$	9.6
${}^{16}\text{O}({}^{16}\text{O}, \gamma){}^{32}\text{S}$	16.5
${}^{16}\text{O}({}^{16}\text{O}, 2\alpha){}^{24}\text{Mg}$	-0.4
${}^{16}\text{O}({}^{16}\text{O}, n){}^{31}\text{S}$	2.1

C burning, $T_c \gtrsim 5 \times 10^8 \text{ K}$



} most probable

O burning, $T_c \gtrsim 1.5 \times 10^9 \text{ K}$



} most probable

Ne & Si burning

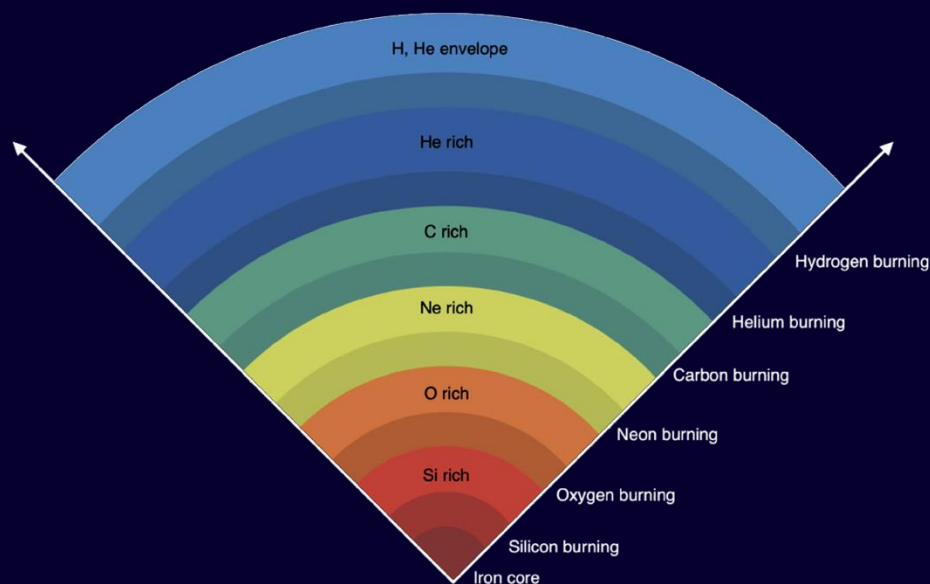


Figure 20 from Arcones & Thielemann 2023 (AARev)

RATE OF PHOTODISINTEGRATION: Define r_f s the rate of forward reactions and λ_r the reverse reaction rate. The temperature at which $r_f \propto \lambda_r$ depends on the Q value of the forward reaction.

$$\lambda_r \propto T_9^{\frac{3}{2}} e^{\left(\frac{-11.6045 Q}{T_9}\right)} r_f$$

NEON BURNING: Photodisintegration of $^{20}\text{Ne}(\gamma, \alpha)^{16}\text{O}$ has a small Q -value of ~ 4 MeV, so reverse rate rivals the forward rate at $\sim 1.5 \times 10^9$ K (preceding O burning).

SILICON BURNING: Partial for $T \sim 3 \times 10^9$ K, complete at $\sim 5 \times 10^9$ K.

QSE: Small groups of nuclei reach equilibrium during the earlier stages of burning in a quasi-nuclear statistical equilibrium. An Si-group and an Fe-group are kept apart by slow reactions with small Q -values.

NSE: As the Si content approaches zero, the remaining reactions also come into equilibrium. All reactions and their isotopes are in a full nuclear statistical equilibrium via both electromagnetic & strong force.

Explosive burning

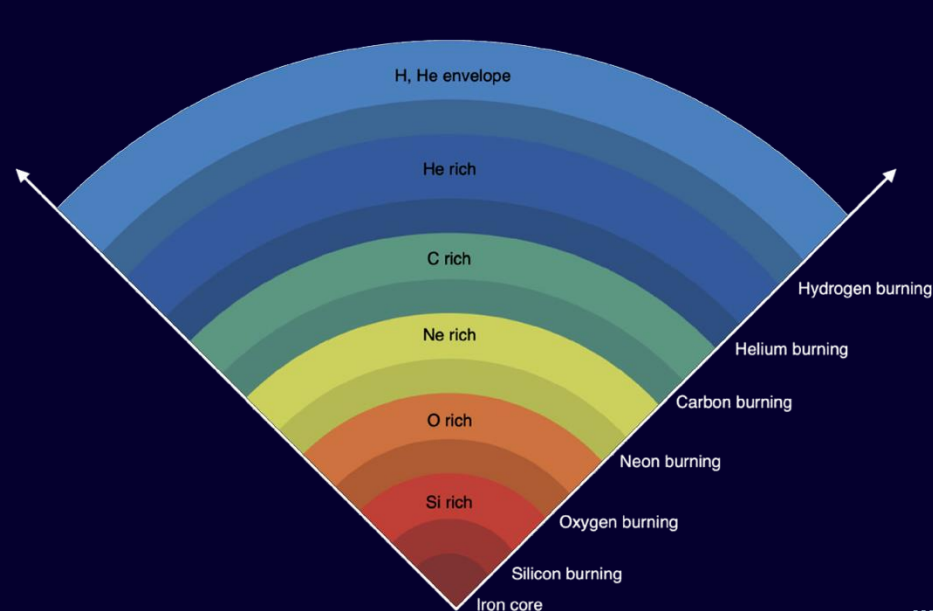
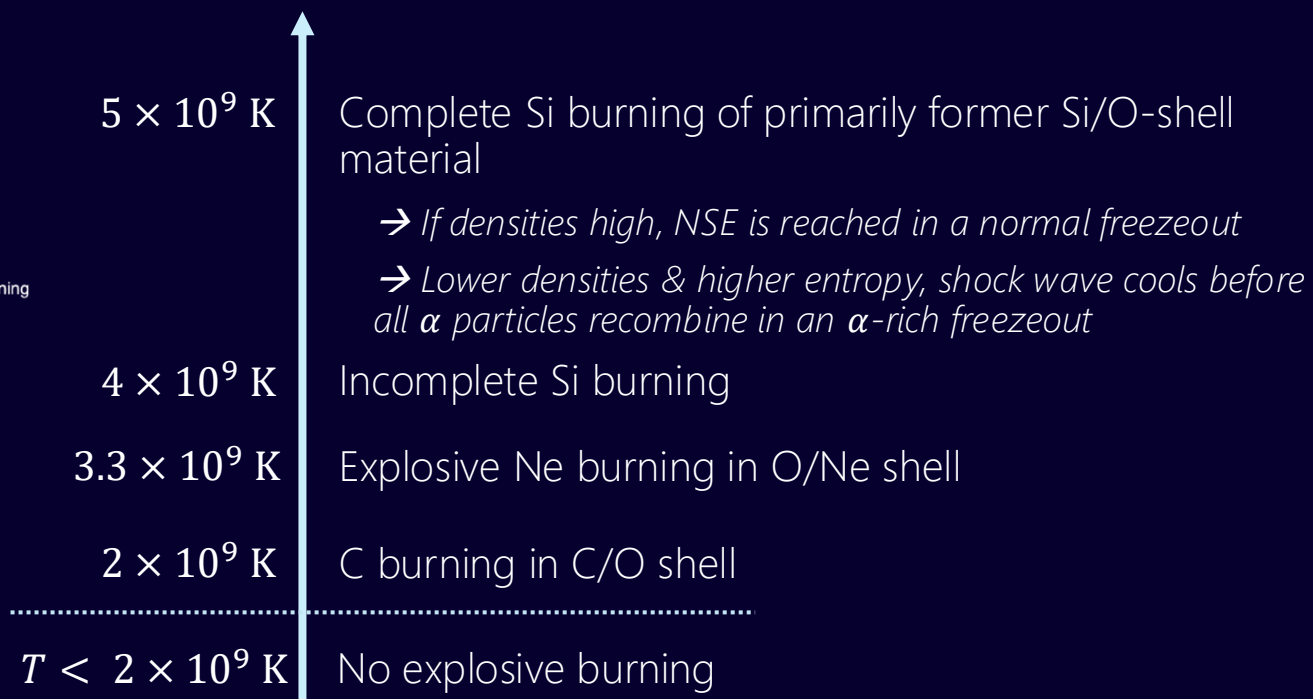


Figure 20 from Arcones & Thielemann 2023 (AARev)



Elements > Fe



Elements > Fe

Neutron Capture Processes

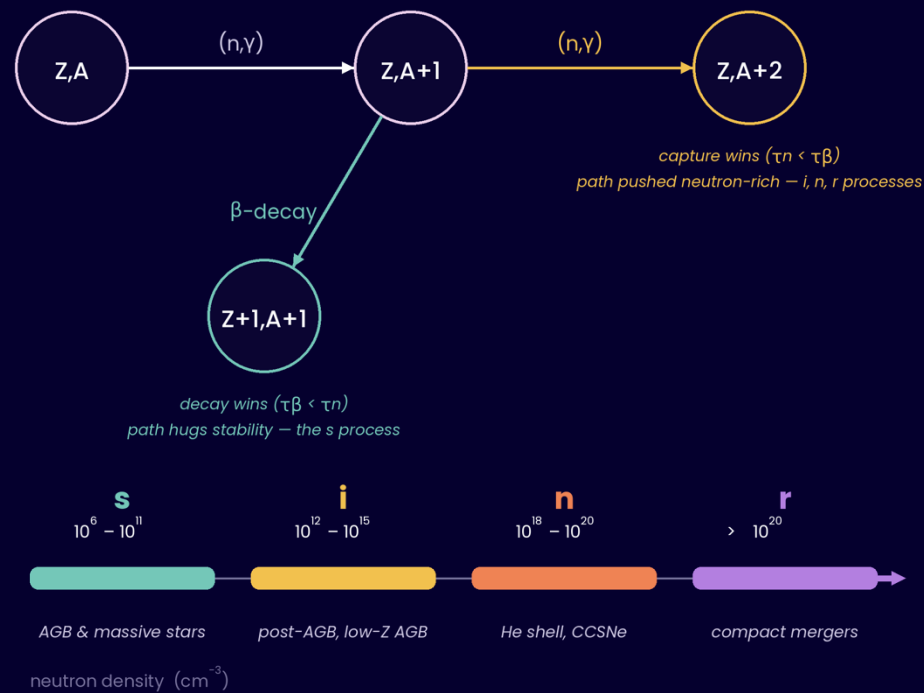


Figure 3 from Arcones & Thielemann 2023 (AARev)

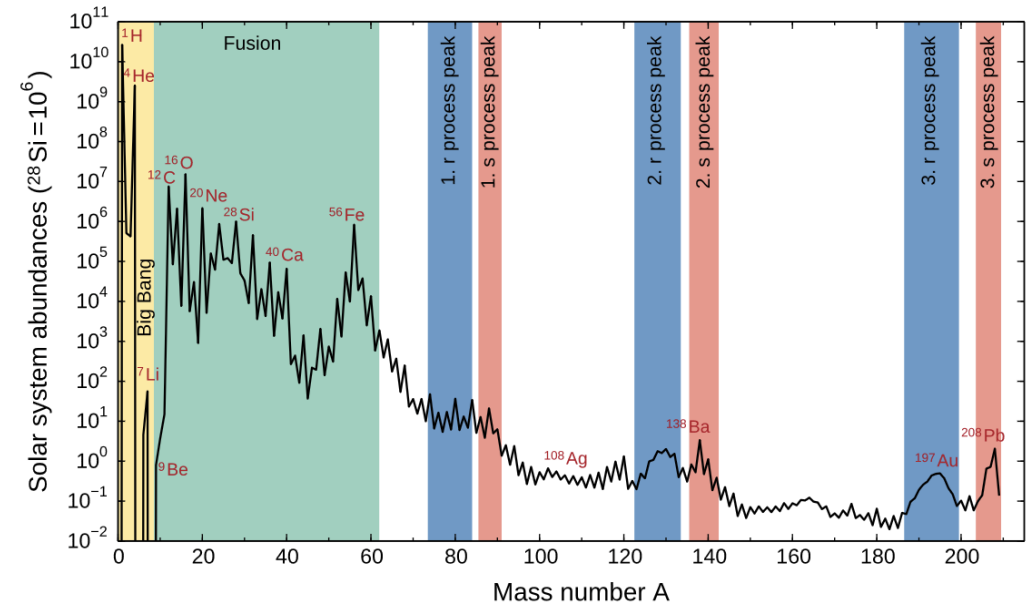
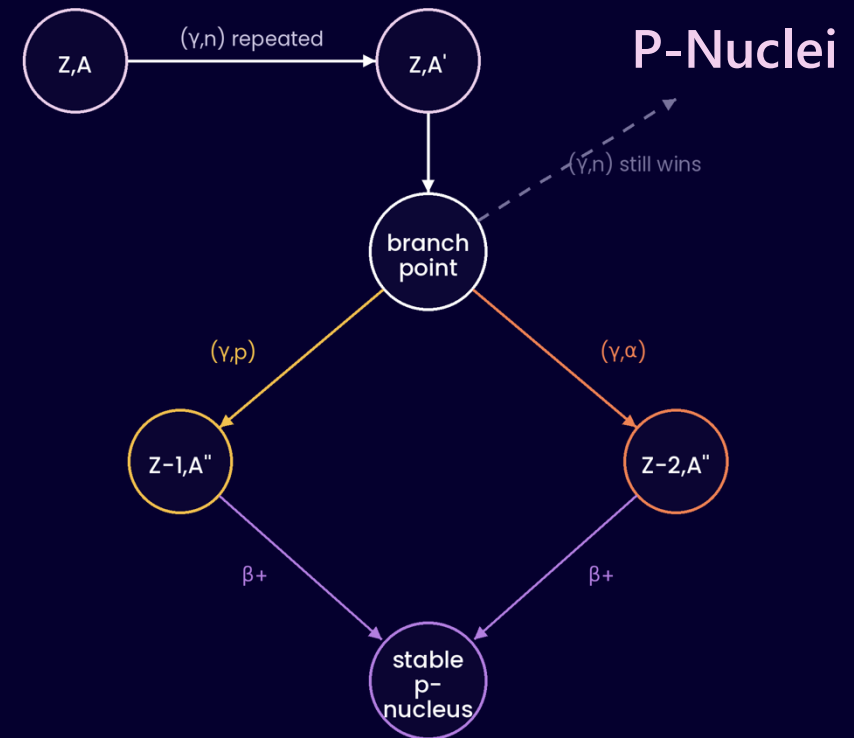
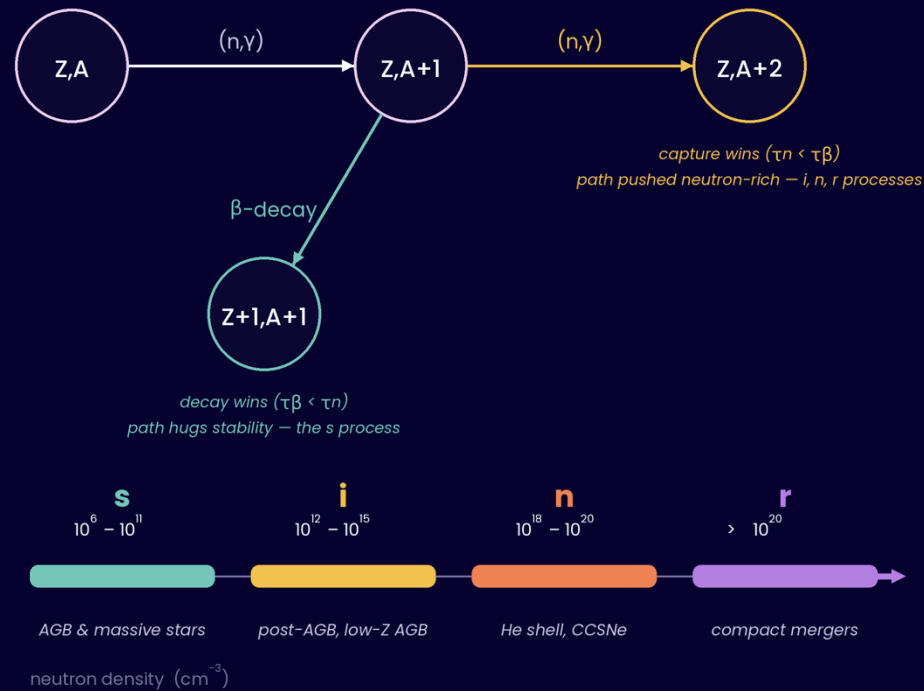


Fig. 3 Solar system abundances based on Lodders (2021) as a function of mass number A ($A = Z + N$), with Z being the number of protons and N the number of neutrons), normalized to Si at 10^6

Elements > Fe

Neutron Capture Processes



Branching between (γ, n) , (γ, p) , and (γ, α) sets which proton-rich isotope survives once decreasing temperature lets $\beta+$ decay carry the flow back to stability.

Elements > Fe

ν p-process

antineutrinos captured on free protons supply neutrons for (n,p) reactions, letting proton captures leap past $N \approx Z$ waiting points

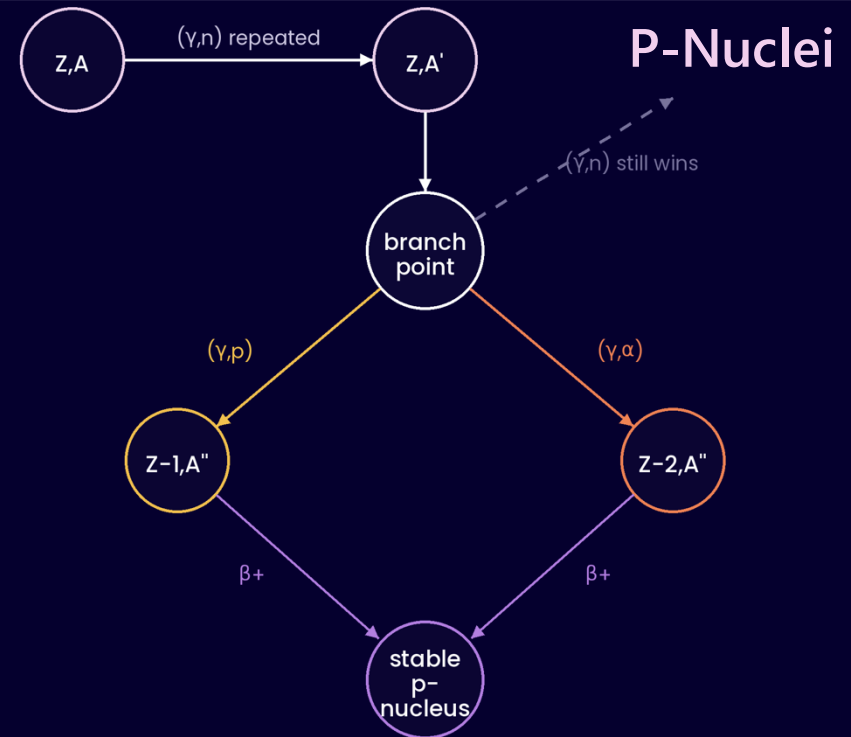
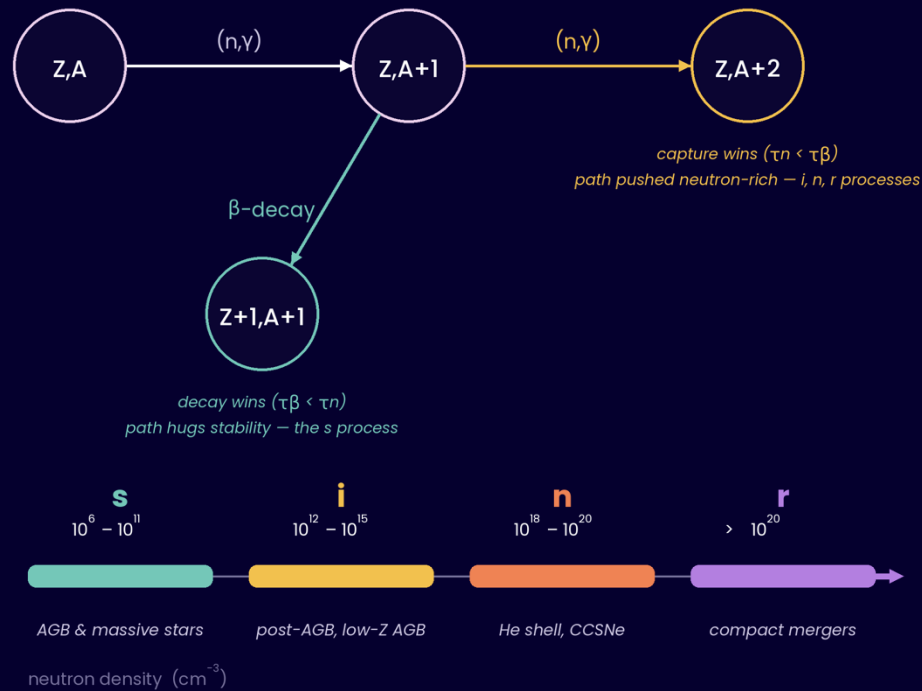
neutrino-driven, proton-rich supernova ejecta

ν -process

neutrinos excite nuclei above the particle-emission threshold; the nucleus sheds a p, n, or α — “neutrino spallation”

stellar mantle, as the supernova shock passes through

Neutron Capture Processes



Branching between (γ, n) , (γ, p) , and (γ, α) sets which proton-rich isotope survives once decreasing temperature lets β^+ decay carry the flow back to stability.

Astrophysical Sites

SINGLE OR BINARIES

AGBs

SNe Type II

Special SNe

BINARIES

NSMs

Type Ia

REGIME: 10s to 100s of periodic cycles. Double-shell burning punctuated by thermonuclear runaways in the He shell (thermal pulses). Pulses drives convective mixing between the processed region and surface (third dredge-up, 3DU). Main site of s-process. H burning can occur at envelope base (hot bottom burning, HBB).

PROGENITOR MASS: Low-mass AGBs undergo deep 3DU, many become C-rich. Dominant neutron source is ^{13}C , producing 2nd- and 3rd-peak species. Intermediate-mass AGBs reach temps for HBB and convert dredged-up ^{12}C to ^{14}N . They are too hot for ^{13}C source, ^{22}Ne activates instead. Lower neutron exposure, production mostly 1st peak.

METALLICITY IMPACT: Higher-Z AGBs are less dense, with smaller He cores at onset. Leads to weaker thermal pulses, greater mass loss, fewer cycles, so 3DU and HBB are both less efficient. The ^{13}C neutron source is primary, so lower $[\text{Fe}/\text{H}]$ means the same neutron supply is shared among fewer Fe seeds pushing flow from 1st peak toward 2nd and 3rd peak.

SIGNATURE PRODUCTS: Low mass AGBs dominant contributors to galactic budget of main s-process component (Sr to Pb) and primary C (through 3DU). Intermediate mass AGBs make primary N and Ne-Na / Mg-Al chains at lower Z, Li at higher Z via Cameron–Fowler mechanism.

REVS: Busso Gallino & Wasserburg (1999, ARAA), Herwig (2005, ARAA), Karakas & Lattanzio (2014, PASA), Lugaro et al. (2023, ARNPS)

Astrophysical Sites

SINGLE OR BINARIES

AGBs

SNe Type II

Special SNe

BINARIES

NSMs

Type Ia

REGIME: Sequential core/shell burning $H \rightarrow He \rightarrow C \rightarrow Ne \rightarrow O \rightarrow Si$ (onion structure), each stage hotter, denser, and shorter-lived than the last. Ends when inert Fe core exceeds Chandrasekhar-like limit and collapses. Rebounding shock can drive explosive nucleosynthesis through shells and reprocessing them.

PROGENITOR MASS IMPACT: Sets how far hydrostatic burning proceeds and how hot each stage runs. At fixed Z_i , higher mass changes core composition to less He, more CNO-processed material, and greater contributions from heavier nuclei. At collapse, the resulting core compactness influences the explosion and the balance between Fe-group fallback and ejecta.

METALLICITY IMPACT: For elements below Fe, higher Z_i favours production of odd- Z elements over α -elements (e.g. Na, Cu) and of more neutron-rich isotopes (e.g. ^{26}Mg over ^{24}Mg). For s-process elements, as a secondary source ^{22}Ne (via $CNO \rightarrow ^{14}N \rightarrow ^{22}Ne$) strengthens with increasing Z , so the weak s-process gets stronger. But when ^{22}Ne is instead primary (built via rotational mixing) the trend reverses and weakens at higher Z (e.g., Roberti et al. 2024 ApJSS).

SIGNATURE PRODUCTS: The dominant Galactic source of the α -elements (O, Ne, Mg, Si, S, Ar, Ca) through both hydrostatic and explosive contributions. The weak s-process (Ga up to 1st peak). Explosive Si burning supplies Fe-group and ^{56}Ni (smaller total than SN Ia).

REVS: Nomoto, Kobayashi & Tominaga (2013, ARAA), Arcones & Thielemann (2023, AARev)

Astrophysical Sites

SINGLE OR BINARIES

| AGBs

| SNe Type II

| **Special SNe**

BINARIES

| NSMs

| Type Ia

MAGNETO-ROTATIONAL SNE:

REGIME: Core collapse of a rapidly rotating, strongly magnetised massive star. Magnetic stresses can drive an early, fast, bipolar/jet-like explosion, augmenting or replacing neutrino heating as the dominant explosion mechanism.

KEY PHYSICS: Rapid magnetic ejection can remove neutron-rich material before neutrino irradiation substantially raises Y_e , providing the high neutron to seed conditions needed for a strong r-process. Nucleosynthetic outcome depends strongly on the magnetic-field strength and configuration, rotation, and neutrino irradiation.

SIGNATURE PRODUCTS: Potentially strong r-process production, reaching the 3rd r-process peak in favourable models. Also produces the usual alpha and Fe-group material and can include proton-rich / vp-process ejecta.

SEE: Reichert et al. (2021 & 2023, MNRAS), Yong et al. (2021, Nature), Nishimura, Takiwaki & Thielemann (2015, Apj)

Astrophysical Sites

SINGLE OR BINARIES

AGBs

SNe Type II

Special SNe

BINARIES

NSMs

Type Ia

PAIR INSTABILITY SNE:

REGIME: Very massive stars with sufficiently massive He cores. At high temperatures, photo-photon pair production reduces radiation pressure, causing the core to contract and heat.

KEY PHYSICS: The pair instability triggers explosive O burning, releasing enough energy to completely disrupt the star rather than producing core collapse. The star is disrupted before the advanced burning and neutronization characteristic of ordinary massive-star evolution can substantially modify the composition, leaving a low neutron excess.

SIGNATURE PRODUCTS: Large yields of alpha and Fe-group elements, substantial Ni56. A pronounced odd-even abundance pattern arising from the low neutron excess.

SEE: Heger & Woosley (2002 & 2005), Kasen et al. (2011, ApJ), Nomoto, Kobayashi & Tominaga (2013, ARAA)

Astrophysical Sites

SINGLE OR BINARIES

■ **AGBs**

■ **SNe Type II**

■ **Special SNe**

BINARIES

■ **NSMs**

■ **Type Ia**

REGIME: Merger of two neutron stars, or a neutron star and black hole. Tidal and shock forces eject neutron-star matter during the merger, while neutrino irradiation and viscous evolution drive additional outflows from the post-merger remnant/accretion disc.

BINARY MASS IMPACT: Determines the merger dynamics and the amount and composition of ejecta. Unequal-mass systems can produce more tidal, highly neutron-rich ejecta, while the total mass and mass ratio influence whether a long-lived neutron-star remnant or prompt black hole forms, and hence the subsequent disc/wind outflows.

METALLICITY IMPACT: Metallicity instead affects the binary progenitor evolution, merger population and delay-time distribution, influencing when and where NSMs enrich galaxies. Chemical-evolution models therefore find that the ability of NSMs to explain early r-process enrichment depends strongly on their delay-time distribution and the properties of the merger population.

SIGNATURE PRODUCTS: Robust heavy r-process production, including the second and third peaks and potentially actinides. Extremely neutron-rich dynamical ejecta favour the heavy r-process, while higher- Y_e neutrino-irradiated and disc outflows can produce lighter r-process nuclei and fewer lanthanides.

REVS: Wanajo et al. (2014, ApJ), Thielemann et al. (2017, ARAA), Just et al. (2015, MNRAS), Metzger (2020, LRR).



Astrophysical Sites

SINGLE OR BINARIES

AGBs

SNe Type II

Special SNe

BINARIES

NSMs

Type Ia

REGIME: Thermonuclear disruption of a degenerate C/O white dwarf through near-Chandrasekhar or sub-Chandrasekhar channels. Burning propagates through the star via deflagration, detonation, or both.

KEY PHYSICS: At $T \gtrsim 7\text{GK}$, C/O burning reaches NSE. Density sets the freeze-out regime: high-density burning produces neutron-rich stable Fe-group nuclei through electron captures; normal freeze-out favours ^{56}Ni , while α -rich freeze-out produces nuclei such as Zn. At $4.5 - 7\text{GK}$, incomplete burning produces Si-S-Ar-Ca.

WD MASS IMPACT: More massive WDs reach higher densities and burn more completely to NSE; lower-mass WDs leave more unburnt intermediate-mass elements. Chandrasekhar-mass models produce more stable neutron-rich Fe-group material relative to ^{56}Ni than sub-Chandrasekhar models; both channels appear necessary to span observed Fe-group ratios.

METALLICITY IMPACT: Acts through ^{22}Ne , scaling the neutron excess with Z . Higher metallicity therefore favours more neutron-rich stable Fe-group isotopes and less ^{56}Ni .

SIGNATURE PRODUCTS: Dominant source of Fe-group elements, including ^{56}Ni , which powers the SN Ia light curve. Also produces intermediate-mass elements (Si, S, Ar, Ca). The balance between ^{56}Ni and stable Fe-group isotopes such as ^{54}Fe , ^{58}Ni , and Mn constrains the explosion density, metallicity, and WD mass.

REVS: Ruiter & Seitenzahl (2025, A&ARv); Keegans et al. (2023, ApJS 268, 8); Sim et al. (2011, IAU Symp. 281); Bulla et al. (2016, MNRAS)

Takeaways

NUCLEOSYNTHESIS IS A PHYSICAL PROBLEM: The elements are set by the interplay of temperature, density, composition, reaction rates, and timescale — determining which reactions can compete and whether material reaches equilibrium.

MASS & METALLICITY MATTER– BUT NOT IN THE SAME WAY EVERYWHERE: Stellar mass and metallicity shape stellar structure, burning conditions, and evolutionary timescales. Metallicity also sets the initial composition, neutron excess and abundance of seed nuclei.

THE HEAVIEST ELEMENTS REQUIRE EXTREME CONDITIONS: Beyond Fe, neutron captures dominate: s-process → sustained neutron exposure; r-process → extreme neutron densities and rapid neutron captures; p-processes → proton-rich or photodisintegration environments.

THE ABUNDANCE OF AN ELEMENT IS THE SUM OF ITS ASTROPHYSICAL ORIGINS: *No single site produces the periodic table. Different sites contribute different isotopes, in different proportions, under different physical conditions.*